Fig. 3.14.

3.8. DUALITY

Consider the two network completely different in physical appearance as shown in Fig. 3.15.

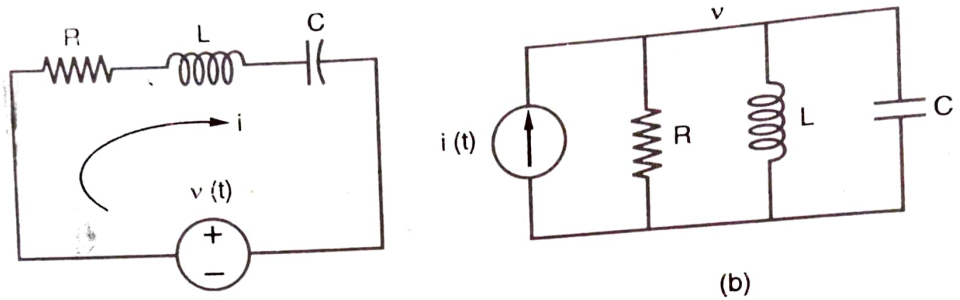


Fig. 3.15. Dual Networks

Fig. 3.15 (*a*) is the series circuit, where KVL is applicable while Fig. 3.15 (*b*) is the parallel circuit and KCL is applicable.

From Fig. 3.15 (*a*), by KVL,

$$v(t) = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt \qquad \ldots(3.15)$$

From Fig. 3.15 (*b*), by KCL,

$$i(t) = \frac{v}{R} + \frac{1}{L}\int v\,dt + C\frac{dv}{dt} \qquad ...(3.16)$$

These two equations specify identical mathematical operations, the only difference being in latter symbols. The solution of two equations are same. Such networks are called *dual networks.*

In the above example quantities which are analogous to each other are called the *dual quantities*. Table 3.1 shows pair of dual quantities.

Table 3.1. Dual Pairs.

Voltage source	Current source
Current (i)	Voltage (v)
Loop	Node pair
Resistance (R)	Conductance (G)
Inductance (L)	Capacitance (C)
Ri	Gv
$L\frac{di}{dt}$	$C\frac{dv}{dt}$
$\frac{1}{C}\int i\,dt$	$\frac{1}{L}\int v\,dt$
short circuit	Open circuit
charge (q)	flux (ψ)
KVL	KCL
Switch open	Switch closed

Procedure for drawing the dual network :

(*i*) Place a node inside each loop and place an extra node for reference or ground.

(*ii*) The nodes are joined through the element of the original network, traversing only one element at a time. For each element traversed in the original network, place the dual element corresponding to the original element.

Fig. 3.16 shows the above procedure for drawing the dual networks.

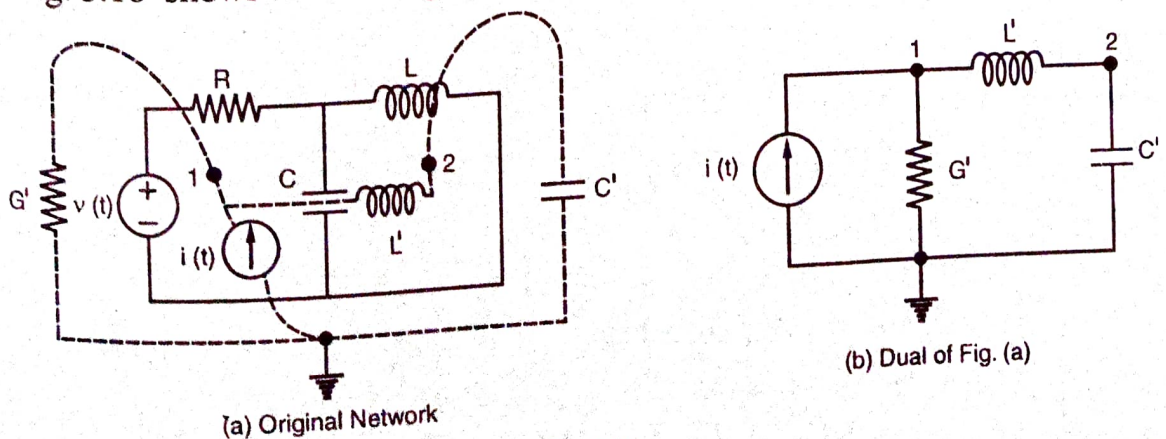


Fig. 3.16.

In the original network of Fig. 3.16 (*a*), there are two loops, one voltage source and three passive elements : R , L and C connected as shown in Fig. 3.16. For drawing the dual network, first we place two nodes (1 and 2) in two loops and one extra node for reference. Now draw the branch between each node in such a way that each branch passes through the one element of original network. Now place the dual element corresponding to the original element in each branch between node 1, 2 and reference node.

Examples on duality :

Example 3.17. *Draw the dual network for the given circuit. (Fig. 3.57.)*

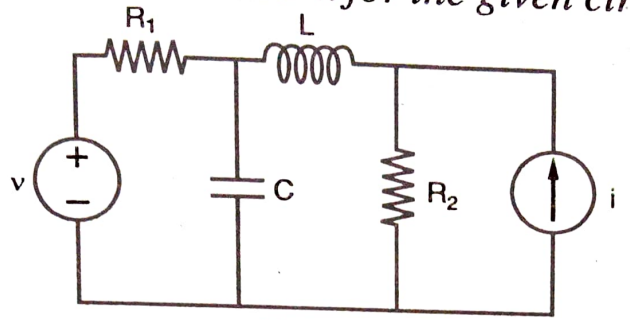


Fig. 3.57.

Solution. Place a node in each loop and one extra node for reference. Draw

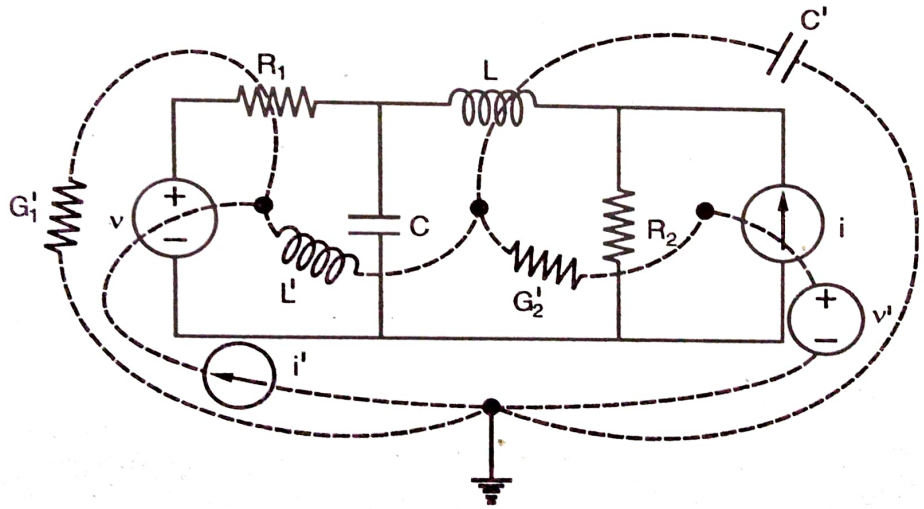


Fig. 3.58.

the branches passing through each element between the nodes. Then place the dual element in each branch corresponding to the original element in the given network.

So the dual network is :

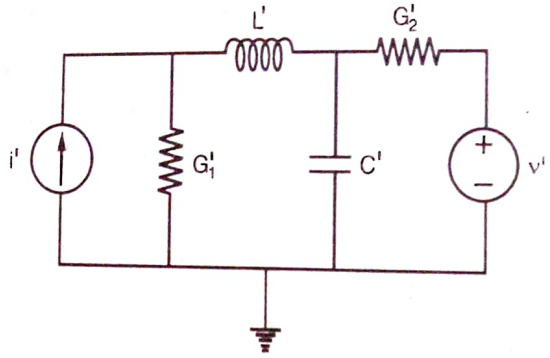


Fig. 3.59.

Example 3.18. *Find the dual of following network.*

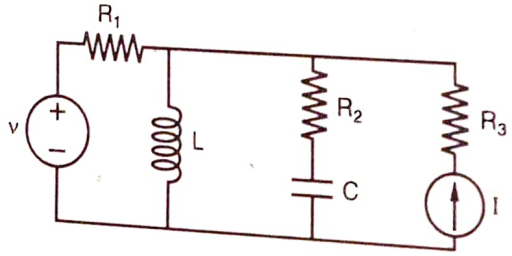


Fig. 3.60.

Solution. Refer Fig. 3.61.

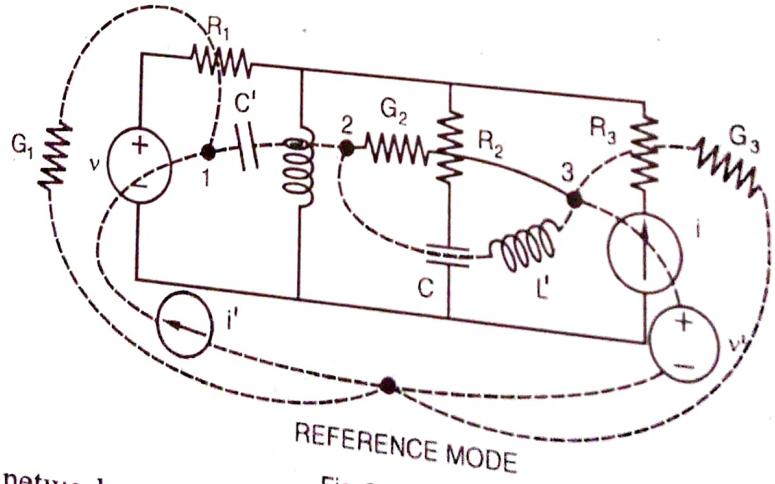


Fig. 3.61.

So the dual network of Fig. 3.60 is

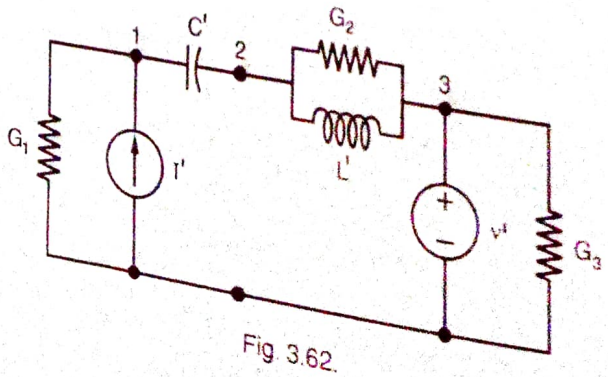


Fig. 3.62.